

# CLAUDE.md

SECTION root    TYPE document    STATUS **active**    OWNER Founder    UPDATED 2026-08-19

Guidance for Claude Code when working in this repository.

## What this is

**ay2m/Office** ("The Office") is the internal **documents repository** for Fly GACA — an independent, educational platform and open regulatory library for Saudi civil aviation. It holds every operating document that runs the company: strategy/OKRs, governance, legal, finance, KSA compliance, HR/people, operations/IT specs, GTM, customer success, investor relations, academy curriculum, and brand — **not** product code. A Claude Code session here is almost always **drafting, editing, or reorganizing organizational documents**, not writing or reviewing application code.

Product code lives in separate repos, all under the **ay2m** account:

| Repo                     |                            | What it is                                                                                                                                                          |
|--------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ay2m/FlyGACA</b>      | private                    | <b>The product.</b> The bilingual web app (React 19 + Vite) <i>and</i> its Express backend on Cloud Run. The regulatory corpus and content pipelines live here too. |
| <b>ay2m/Captain-Adel</b> | private                    | The AI flight instructor service behind captadel.com                                                                                                                |
| <b>ay2m/FlyGACA-ios</b>  | public                     | The native SwiftUI family — one shared package, one App Store app per exam module (ELPT, AIP)                                                                       |
| <b>ay2m/FlyGACA-app</b>  | public,<br><b>archived</b> | The retired predecessor of <b>ay2m/FlyGACA</b> . Read-only, kept for its 1,005-commit history. Do not cite it as current.                                           |

[!NOTE] Older documents in this tree refer to a **FlyGACA/...** org and to six per-module App Store metadata repos ( **PPL** , **CPL** , **IR** , **ATPL** , **ELPT** , **AIP** ). **Neither is current.** The **FlyGACA/...** paths are legacy redirects to **ay2m/...** , and those six repos do not exist. Some also call **ay2m/FlyGACA** "the iOS family" — it is the web app and backend; iOS is **ay2m/FlyGACA-ios** .

[!IMPORTANT] Fly GACA is **not affiliated with GACA** (Saudi General Authority of Civil Aviation). This constraint is load-bearing across the whole document tree, not just user-facing product copy — legal, GTM, and investor materials must all state the relationship accurately.

This repo contains real operating material — signed/draft legal agreements, financial data, HR records, investor materials, strategic plans. Treat its contents as sensitive: don't summarize or quote its financial/legal/HR/investor content into other contexts (chat replies, other repos, external tools) beyond what the task actually requires, and never publish it. Security or data-isolation concerns go to the maintainer directly per `01-governance/SECURITY.md` — not a public issue.

## Repository structure

Twelve numbered sections, each independently browsable:

| #                                   | Section              | Contents                                                                                                                                                                              |
|-------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>00-strategy/</code>           | Strategy             | Annual plan & OKRs, master roadmap, CEO execution plan, Phase 0 tracker, numbered brainstorm                                                                                          |
| <code>01-governance/</code>         | Governance           | Founders' agreement, SHA, ESOP, Code of Conduct, board pack, decision log, <code>CLAUDE.md</code> (see below), <code>CONTRIBUTING.md</code> , <code>SECURITY.md</code>                |
| <code>02-legal/</code>              | Legal                | NDA templates, EULA, SLA, pilot agreement, PDPL policies, DPA, IP/takedown procedure                                                                                                  |
| <code>03-finance/</code>            | Finance              | Banking policy, procurement, expense policy, budget-vs-actual tracker, KPI dashboard, ZATCA invoicing (HTML tax-invoice + VAT-return templates)                                       |
| <code>04-compliance-ksa/</code>     | Compliance (KSA)     | MISA license, ZATCA e-invoicing, PDPL DPIA, BCP/DR, Nitaqat plan                                                                                                                      |
| <code>05-people/</code>             | People               | Employment contracts, employee handbook, offer letters, onboarding/offboarding, HR policies                                                                                           |
| <code>06-operations-it/</code>      | Operations/IT        | Digital office setup, product specs (CRM, Captain Adel, Instructor Dashboard), hosting/secrets placement facts, <code>runbooks/</code> , <code>diagrams/</code> , <code>setup/</code> |
| <code>07-gtm/</code>                | Go-To-Market         | Sales playbook, demo script, objection handling, cold outreach, B2B pipeline, <code>seo/</code> strategy                                                                              |
| <code>08-customer-success/</code>   | Customer Success     | Onboarding playbook, health scoring, NPS, QBR templates, at-risk/expansion playbooks                                                                                                  |
| <code>09-investor-relations/</code> | Investor Relations   | Pitch deck, FAQ, due-diligence questionnaire, investor update template, risk register                                                                                                 |
| <code>10-academy-curriculum/</code> | Academy & Curriculum | Curriculum map, coverage matrix, PPL mock exams, B2C learner paths, instructor onboarding                                                                                             |
| <code>11-brand/</code>              | Brand                | Design system, design tokens, style guide (Falcon Theme), logos, print assets (EN + RTL AR)                                                                                           |

Support directories:

- `templates/` — reusable `.md` starters (`tpl-fin-report.md`, `tpl-hr-policy.md`, `tpl-legal-memo.md`, `tpl-ops-runbook.md`, `tpl-strat-proposal.md`; mirrored under `ar/templates/`). Base

new docs on these.

- `ar/` — a parallel Arabic (Saudi MSA) mirror of the **Markdown layer** across all 12 sections — same folder structure and filenames, translated content (118 `.md`, near-parity with the English tree's 119). The `.docx` / `.xlsx` / `.html` deliverables are EN-only. **English is authoritative** — on any conflict the English tree governs. Filenames stay ASCII kebab-case even under `ar/` for easy diffing.
- `tools/print/` — the Markdown → branded A4 PDF pipeline (see below).
- `_print/` — generated PDFs, mirroring the whole tree (including `ar/`). Generated output, but **is committed** (not gitignored) — the CI gate below checks it's present and fresh.
- `_INDEX.md` / `ar/_INDEX.md` — master index of the whole tree.
- `ar/_GLOSSARY.md` — the EN↔AR terminology glossary that keeps translations consistent.

Several document formats coexist: polished deliverables are committed binaries

( `.docx` / `.xlsx` / `.pptx` , plus source PDFs, investor-deck JPGs, and brand PSD/PNG assets — SVG logos/diagrams are committed as text — with `.gitattributes` declaring the binary set `.docx` / `.xlsx` / `.pptx` / `.pdf` / `.psd` / `.png` / `.jpg` / `.jpeg` / `.gsheet` ); working notes, specs, drafts, and playbooks are `.md` — which is what the print pipeline and the doc-check CI both operate on.

**18 pages are authored as HTML instead of Markdown** and render through `build-html.mjs` (below): the two bilingual investor decks in `09-investor-relations/decks/`, the four showcase pages ( `00-strategy/brainstorms/00-strategic-brainstorms-dashboard.html` , `00-strategy/the-book-of-fly-gaca.html` , `11-brand/design-system.html` , `11-brand/tidal-reckoning.html` ), the two ZATCA finance templates in `03-finance/` ( `tax-invoice-template.html` , `vat-return-worksheet.html` ), and the ten print-collateral sources in `11-brand/print/` (letterheads EN/AR, memo, press release, business cards, envelope, compliments slip, contract cover — these share `11-brand/print/brand-print.css` ; the showcase/finance pages are self-contained).

`tools/print/build-png.mjs` additionally re-screenshots the `11-brand/print/` sources into the 300 dpi PNGs the brand catalogue references.

## The doc convention (enforced by CI)

Every content `.md` file requires YAML front-matter with these exact keys:

```
---
title: <doc title>
section: <e.g. 03-finance>
doc_type: <document|readme|template|...>
status: <active|draft|scaffold>
owner: <e.g. Founder>
last_updated: <YYYY-MM-DD>
lang: <en|ar>
---
```

Exemptions: the root `README.md` skips only the **front-matter** check — its PDF must still exist under `_print/` and stay fresh, like any other `.md` ; the checker never descends into `tools/**` or `_print/**` at all; files under any `templates/` directory still need a front-matter block but only the

`title` key inside it (they use their own authoring schema, not the full doc metadata). Every other `.md`, anywhere in the tree (including `ar/`), needs the full set.

`status: draft` or `status: scaffold` stamps an automatic DRAFT/SCAFFOLD watermark on the rendered PDF — set it deliberately. (No doc currently uses `scaffold`; the tooling supports it.)

Every `.md` must have a matching, up-to-date PDF under `_print/`. After creating or editing any `.md` file (English or Arabic), regenerate its PDF before committing:

```
cd tools/print
npm ci                    # once, or after a dependency change (no browser download –
uses the                  # pre-installed Chromium the print pipeline auto-detects)
                           # incremental – only content-changed docs re-render
npm run build              # rebuild every doc from scratch
npm run build:force        # render one file
node build.mjs <path/to/doc.md>
node build-html.mjs        # separately renders ALL 16 HTML pages to their _print/
PDFs
node build-png.mjs         # re-screenshots 11-brand/print/*.html to the 300dpi
catalogue PNGs
node check.mjs             # the CI gate itself – front-matter + Markdown-PDF
coverage +                 # staleness + HTML-PDF coverage; no browser needed, safe
anytime
```

(`tools/print/package.json` also exposes `npm run build:html` and `npm run check` aliases.) Adding or renaming **any** `.html` in the tree without running `build-html.mjs` fails CI, same as an unrebuilt `.md`.

`build.mjs` hashes each source file (folded with a hash of `theme.css` + `build.mjs` itself) into `tools/print/.buildcache.json`; `check.mjs` recomputes that hash and fails if it doesn't match the committed PDF, i.e. if you edited a doc but didn't rebuild its PDF. **Commit the regenerated PDF and the updated `.buildcache.json` together with the `.md` change** — a doc edit without a rebuilt PDF fails CI (`.github/workflows/docs-check.yml` — the only workflow — runs on push to `main` and on PRs touching `**/*.md`, `**/*.html`, `_print/**`, `tools/print/**`, or the workflow file itself; CI runs Node 20).

[!WARNING] Editing `build.mjs` marks every PDF in the repo stale, because its own bytes are folded into the shared `themeHash`. For a change that genuinely alters rendering, that's correct — rebuild. But for a change that *cannot* alter output (adding a `SKIP_DIRS` entry, a comment, a log line), re-rendering 238 PDFs is pure churn: Chromium restamps `CreationDate` on every one, so all 238 show up as modified binaries. In that case re-stamp `.buildcache.json` with the new `themeHash` instead — recompute `sha256(themeHash + source)` per existing key and write it back with `JSON.stringify(cache, null, 1) + '\n'`. Verify with `node check.mjs` and a clean `git status _print/`.

Styling is the "Falcon Theme" documented in `11-brand/fly-gaca-document-style-guide.md`: Inter body · Cairo headings (and all Arabic text) · JetBrains Mono code, A4 page, 0.75in margins (0.9in bottom for the footer), footer page numbers, cover block generated from the front-matter. Fonts

are vendored under `tools/print/fonts/` so the whole pipeline runs offline; it auto-detects Chromium via `$PLAYWRIGHT_BROWSERS_PATH` (default `/opt/pw-browsers`) or `$CHROMIUM_PATH`. Requires Node 18+ and **Chromium ≥ 131** (CSS `@page` margin boxes drive the footer page numbers).

## Conventions

- **Bilingual, English-authoritative.** New or materially changed content docs should eventually get an `ar/` counterpart at the same path, translated per `ar/_GLOSSARY.md`'s terminology — but unlike the product repos, CI here does **not** hard-gate on EN/AR parity (only front-matter + PDF freshness are checked independently per file), so an EN-only doc won't fail CI. Still keep the mirror in sync as a matter of practice; don't let it drift silently for long.
- **Sensitive-content discipline.** Legal, financial, HR, and investor documents in this repo are real operating material, not samples — quote or restate the minimum necessary for the task.
- **Governance is the source of policy**, not this file. `01-governance/CONTRIBUTING.md` is now a repo-native guide (rewritten 2026-08-09 — it used to be a vendored copy of the application repo's guide; its `npm run check:*` commands never applied here). `01-governance/SECURITY.md` is filled in and current — for security or data-isolation concerns email the maintainer directly (address in `SECURITY.md`). `01-governance/CODE_OF_CONDUCT.md` is real. `01-governance/company-facts.md` is the canonical entity-facts doc the *product* repos consume — check it before restating company facts anywhere.
- `01-governance/CLAUDE.md` used to be a stale vendored copy of a *different* repo's Claude guidance (the product monorepo's, describing `functions/`, routing, etc. — none of which exists here); it has been replaced with a short pointer back to this file. If you find it drifting again, fix the pointer, don't restore old content.
- License: Apache 2.0 (`01-governance/LICENSE`) for the repo's own material; regulatory content quoted anywhere belongs to GACA and is not covered by that license.

## Where to look

📖 **Family context:** [The Book of Fly GACA](#) is the whole-family reference — all ten repos, the shared tenets, and the glossary in one place.

- `_INDEX.md` — the readable master index across all 12 sections (and `ar/_INDEX.md` for the mirror). Note it defers to `00-strategy/00-master-office-paperwork-index.gsheet` as the authoritative master index.
- `00-strategy/roadmap.md` — current phase status and what's next.
- `tools/print/README.md` — full print-pipeline usage/config detail beyond the summary above.
- `01-governance/` — `company-facts.md`, `CODE_OF_CONDUCT.md`, `decision-log.md`, `CONTRIBUTING.md`, `SECURITY.md`.
- `ar/_GLOSSARY.md` — EN↔AR terminology, use it before translating anything.